

Representations as sums of distinct terms $k/2^k$

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Abstract

For $f(k) := k/2^k$ with $k \in \mathbb{Z}_{\geq 1}$, we study three questions about representing numbers as sums of *distinct* terms $f(a_k)$ with mutually distinct integers $a_k \geq 1$. We prove unconditionally that infinitely many n satisfy $n/2^n = \sum_{k \leq t} f(a_k)$ with $t \geq 2$ (Theorem 3.1), give an exact graph-theoretic reformulation of the all- n question (Proposition 4.2), verify it for every $n \leq 2000$ and for an explicit infinite family (Section 4), and leave the general all- n statement as a precise conjecture equivalent to reachability in a corridor digraph. For continuum-many infinite representations, we prove that $x = 2$ has $2^{\aleph_0}$ solutions when “solution” means an injective sequence (a_k) (Theorem 5.2), and we construct an explicit irrational x^* with $2^{\aleph_0}$ distinct set-solutions (Theorem 5.3). The rational-set case is shown to be exactly as hard as the all- n finite problem. An appendix documents exact-arithmetic verification code.

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1 Statement of the problem and summary of results

Throughout, write

$$f(k) := \frac{k}{2^k}, \quad k \in \mathbb{Z}_{\geq 1}.$$

The three questions are:

- (1) Are there infinitely many n such that, for some $t \geq 2$ and *mutually distinct* integers $a_1, \dots, a_t \geq 1$,

$$\frac{n}{2^n} = \sum_{k=1}^t \frac{a_k}{2^{a_k}}?$$

- (2) Is the same property true for *every* n ?

- (3) Is there a rational x such that

$$x = \sum_{k=1}^{\infty} \frac{a_k}{2^{a_k}}$$

has at least $2^{\aleph_0}$ solutions (with the a_k mutually distinct)?

The requirement that the a_k be mutually distinct is essential: without it the problem is trivial (e.g. repeated $f(2)$).

Summary.

Question	Answer	Status
(1) Infinitely many n ?	Yes	Theorem 3.1
(2) True for all n ?	Yes for $n \leq 2000$ and an infinite family; conjectured in general	Section 4 ; Open Problem 4.2
(3) Rational x with $2^{\aleph_0}$ solutions?	Yes (sequence reading); rational <i>set</i> -reading open	Theorems 5.2 , 5.3 ; Section 5.4

2 Preliminaries

Lemma 2.1 (Telescoping). *Let $G(m) := (m+1)/2^{m-1}$ for $m \geq 1$. Then*

$$G(m) - G(m+1) = \frac{m}{2^m} = f(m),$$

and hence

$$\sum_{k \geq m} f(k) = G(m) = \frac{m+1}{2^{m-1}}, \quad (1)$$

$$\sum_{k=m}^M f(k) = \frac{m+1}{2^{m-1}} - \frac{M+2}{2^M}. \quad (2)$$

In particular $\sum_{k \geq 1} f(k) = 2$.

Proof. Direct algebra: $G(m) - G(m+1) = \frac{m+1}{2^{m-1}} - \frac{m+2}{2^m} = \frac{2(m+1)-(m+2)}{2^m} = m/2^m$. $\square$

Remark 2.2 (Indices must exceed n when $n \geq 3$). For $n \geq 3$, any solution of $n/2^n = \sum_k f(a_k)$ with distinct a_k satisfies $a_k > n$ for all k . Indeed f is strictly decreasing for $k \geq 2$, and $f(n-1) = \frac{2n-2}{2^n} > \frac{n}{2^n}$; including any index $\leq n-1$ overshoots the target, while $a_k = n$ would force $t = 1$. For $n = 1, 2$ one checks directly that every index in a solution with $t \geq 2$ is at least 3.

3 Question 1: infinitely many n

Theorem 3.1 (Explicit infinite family). *For every integer $s \geq 3$, set $n = n_s := 2^s - s - 1$. Then*

$$\boxed{\frac{n_s}{2^{n_s}} = \sum_{k=n_s+1}^{2^s-2} \frac{k}{2^k}}$$

with $t = s - 1 \geq 2$ distinct terms. Hence there are infinitely many n as in Question (1).

Proof. By (2) with $m = n + 1$ and $M = 2^s - 2$,

$$\sum_{k=n+1}^M f(k) = \frac{n+2}{2^n} - \frac{M+2}{2^M}.$$

The desired identity is therefore equivalent to $M+2 = 2^{M-n+1}$. With $M = 2^s - 2$ and $n = 2^s - s - 1$ we have $M - n + 1 = s$ and $M + 2 = 2^s$, which holds. The indices $n + 1, \dots, 2^s - 2$ are distinct and the number of terms is $M - n = s - 1 \geq 2$ for $s \geq 3$. Distinct values of s give distinct n_s . $\square$

Remark 3.2 (First instances).

$$\frac{4}{2^4} = \frac{5}{2^5} + \frac{6}{2^6}, \quad \frac{11}{2^{11}} = \frac{12}{2^{12}} + \frac{13}{2^{13}} + \frac{14}{2^{14}},$$

$$\frac{26}{2^{26}} = \sum_{k=27}^{30} \frac{k}{2^k}, \quad \frac{57}{2^{57}} = \sum_{k=58}^{62} \frac{k}{2^k}, \dots$$

Remark 3.3 (A second telescoping family). For every odd $E \geq 5$, define

$$b := \frac{2^E - 5}{3} + 1, \quad a := b + 1 - E.$$

Then $a, b \in \mathbb{Z}_{\geq 1}$ and

$$f(a) + f(b) = \sum_{k=a+1}^{b-1} f(k).$$

Proof. By (2) the right-hand side equals $(a+2)2^{-a} - (b+1)2^{1-b}$. Equating with $f(a) + f(b)$ and clearing denominators yields $3b + 2 = 2^{b-a+1}$. Since $b - a + 1 = E$, this is $3b + 2 = 2^E$, which holds by definition of b . Odd E ensures $b \in \mathbb{Z}$. First instances: $(a, b) = (6, 10), (36, 42), (162, 170), (672, 682)$.

4 Question 2: all n ?

4.1 Graph-theoretic reformulation

Definition 4.1 (Corridor digraph). For a target n , a finite representation $S \subset \mathbb{Z}_{\geq 1}$ corresponds to a walk in the digraph Γ_n whose vertices at *scale* $j \geq n$ are integers ρ with $0 \leq \rho \leq j + 2$, and whose edges (when the target lies in the corridor) are

$$(j, \rho) \longrightarrow (j + 1, 2\rho) \quad (\text{exclude } j + 1), \quad (j, \rho) \longrightarrow (j + 1, 2\rho - (j + 1)) \quad (\text{include } j + 1).$$

Proposition 4.2 (Exact equivalence). *Fix $n \geq 1$. There exist mutually distinct integers $a_1, \dots, a_t \geq 1$ with $t \geq 1$ and*

$$\frac{n}{2^n} = \sum_{k=1}^t f(a_k)$$

if and only if the vertex (n, n) of Γ_n can reach some vertex $(J, 0)$ with $J \geq n$. In that case one may take $S = \{k \in (n, J] : \text{the edge at scale } k-1 \text{ is an Include edge}\}$.

Proof. Define the remainder at scale $j \geq n$ by

$$\rho_j := 2^j \left(\frac{n}{2^n} - \sum_{k \in S, k \leq j} f(k) \right),$$

with S the index set being built. Then $\rho_n = n$, each $\rho_j \in \mathbb{Z}$, and the transition rule is

$$\rho_{j+1} = 2\rho_j - (j+1) \mathbf{1}[j+1 \in S],$$

exactly the two edge types above. By (1), the total tail after scale j is $2^{-j}(j+2)$, so any finite representation stays in $0 \leq \rho_j \leq j+2$. Conversely, reaching $(J, 0)$ means the partial sum equals $n/2^n$ exactly with all chosen indices $\leq J$. Distinctness of indices is built into using each scale at most once. $\square$

Remark 4.3 (Branching is rare). Both edges leave (j, ρ) only when $2\rho \in \{j+1, j+2, j+3\}$, a window of width 3 inside a corridor of width $j+3$. The walk is therefore almost deterministic; hitting $\rho = 0$ is a delicate exact problem.

4.2 Verified instances and computational theorem

Theorem 4.4 (Certificates for small n). *For each n in Table 1, the listed set S consists of distinct integers ≥ 1 with $t = |S| \geq 2$ and $\sum_{k \in S} f(k) = n/2^n$. All entries were verified by exact rational arithmetic.*

Table 1: Explicit certificates with $t \geq 2$.

n	a solution S
1	$\{4, 5, 6\}$
2	$\{4, 5, 6\}$
3	$\{4, 6, 8\}$
4	$\{5, 6\}$ (Theorem 3.1, $s = 3$)
5	$\{6, 7, 11, 13, 14\}$
6	$\{7, 8, 11, 13, 14\}$
7	$\{8, 9, 12, 13, 14, 15, 20, 21, 24\}$
9	$\{10, 11, 13, 14\}$
11	$\{12, 13, 14\}$ (Theorem 3.1, $s = 4$)
12	$\{13, 14, 15, 20, 21, 24\}$
15	$\{16, 17, 18, 21, 22\}$

Theorem 4.5 (Machine-verified range). *For every integer n with $1 \leq n \leq 2000$, the vertex (n, n) reaches $(J, 0)$ in Γ_n for some $J \leq 250000$. Hence Question (2) has answer **yes** for all $n \leq 2000$.*

Proof. Appendix A gives a forward reachability program using exact integer arithmetic only. Running it for $n = 1, \dots, 2000$ finds a hitting scale $J \leq 250000$ in every case. Each hit yields an explicit Include/Exclude certificate; summing the included terms gives $n/2^n$ by Proposition 4.2. $\square$

Together with Theorem 3.1, Question (2) is resolved for an infinite set of n and for all $n \leq 2000$.

Open problem (All n). For every $n \geq 1$, does (n, n) reach 0 in Γ_n ? Equivalently: does every n admit a representation $n/2^n = \sum_{k \leq t} f(a_k)$ with $t \geq 2$ and distinct a_k ?

Remark 4.6 (Why a complete proof is hard). The coin system $\{f(k)\}$ sits at the *critical* threshold for subset-sum completeness: $\sum_{j > k} f(j)/f(k) = (k+2)/(2k) \rightarrow \frac{1}{2}$, so there is asymptotically no slack. Branch vertices in Γ_n have $O(1)$ width in a corridor of width $\Theta(j)$; proving that the exact value 0 is always hit is a global hitting-time statement that resisted every elementary block-telescoping mechanism we tried (such mechanisms collapse to the rigid power-of-two condition $M+2 = 2^{M-n+1}$ underlying Theorem 3.1).

5 Question 3: continuum-many infinite representations

5.1 Completeness of infinite sums

Lemma 5.1 (Greedy completeness). *Every $x \in (0, 2]$ equals $\sum_{k \in S} f(k)$ for some set $S \subseteq \mathbb{Z}_{\geq 1}$ of distinct indices (possibly infinite).*

Proof. Process $k = 1, 2, 3, \dots$; maintain remainder r , initially $r = x$. Include k iff $f(k) \leq r$. Invariant before step k : $0 \leq r \leq G(k) = \sum_{j \geq k} f(j)$ (true at $k = 1$ since $x \leq 2 = G(1)$). If k is included, $r \leftarrow r - f(k) \leq G(k) - f(k) = G(k+1)$. If k is skipped, $r < f(k) \leq G(k+1)$ because $f(k) \leq G(k+1)$ is equivalent to $k \leq k+2$. Thus the invariant persists, $r \rightarrow 0$, and the selected sum converges to x . $\square$

5.2 Reading A: solutions as sequences

If a “solution” of $x = \sum_{k=1}^{\infty} a_k/2^{a_k}$ is an injective sequence $(a_k)_{k \geq 1}$, then:

Theorem 5.2 (Rational $x = 2$). *The rational number $x = 2$ has at least $2^{\aleph_0}$ distinct solutions.*

Proof. For $\varepsilon \in \{0, 1\}^{\mathbb{N}}$, let π_ε be the permutation of $\mathbb{Z}_{\geq 1}$ that transposes $2i-1$ and $2i$ iff $\varepsilon_i = 1$. Set $a_k := \pi_\varepsilon(k)$. Then (a_k) is injective with image $\mathbb{Z}_{\geq 1}$. Nonnegative series have order-independent sums, so

$$\sum_{k=1}^{\infty} f(a_k) = \sum_{m \geq 1} f(m) = 2$$

by Lemma 2.1. If $\varepsilon \neq \varepsilon'$, they differ at some i , hence the sequences differ at index $2i-1$. This gives $2^{\aleph_0}$ distinct solutions. $\square$

5.3 Reading B: solutions as sets

If a “solution” is an infinite set $A \subset \mathbb{Z}_{\geq 1}$ with $\sum_{k \in A} f(k) = x$ (rearrangements not counted separately), then:

Theorem 5.3 (Continuum-many set-solutions). *Let $n_s = 2^s - s - 1$ and $B_s = \{n_s + 1, \dots, 2^s - 2\}$ as in Theorem 3.1, and define*

$$x^* := \sum_{s \geq 3} \frac{n_s}{2^{n_s}} \in (0, 2).$$

Then the equation $x^ = \sum_{k \in A} f(k)$ has at least $2^{\aleph_0}$ pairwise distinct infinite sets A .*

Proof. The intervals $I_s := [n_s, 2^s - 2]$, $s \geq 3$, are pairwise disjoint because

$$\min I_{s+1} - \max I_s = (2^{s+1} - s - 2) - (2^s - 2) = 2^s - s > 0.$$

For $\varepsilon \in \{0, 1\}^{\{3, 4, 5, \dots\}}$ define

$$A_\varepsilon := \bigcup_{s \geq 3} X_s(\varepsilon_s), \quad X_s(0) := \{n_s\}, \quad X_s(1) := B_s.$$

Each A_ε is an infinite set of distinct integers. Since all terms are positive,

$$\sum_{k \in A_\varepsilon} f(k) = \sum_{s \geq 3} \sum_{k \in X_s(\varepsilon_s)} f(k) = \sum_{s \geq 3} f(n_s) = x^*,$$

using Theorem 3.1 at each s . The map $\varepsilon \mapsto A_\varepsilon$ is injective because $n_s \in A_\varepsilon$ iff $\varepsilon_s = 0$. $\square$

Proposition 5.4 (Irrationality of x^*). *The number x^* is irrational.*

Proof. Each summand $n_s/2^{n_s}$ contributes a block of 1-bits in the binary expansion occupying positions roughly $[n_s - \lfloor \log_2 n_s \rfloor, n_s]$, and these blocks lie in the disjoint intervals $I'_s = [n_s - \lfloor \log_2 n_s \rfloor, n_s]$. Successive blocks are separated by gaps of length at least $2^s - s - \lfloor \log_2 n_{s+1} \rfloor \rightarrow \infty$. Because the blocks are disjoint there are no carries between them, so the genuine binary expansion of x^* contains infinitely many isolated 1-bits with unbounded gaps of 0s between them. An eventually periodic dyadic expansion cannot have both properties: if the period contains a 1, gaps between 1s are bounded; if not, there are only finitely many 1s. Hence x^* is not rational. $\square$

5.4 Rational x in the set reading

Remark 5.5 (Dichotomy via Cantor–Bendixson). For fixed x , the family $\mathcal{F}_x := \{A \subseteq \mathbb{Z}_{\geq 1} : \sum_A f = x\}$ is closed in $\{0, 1\}^{\mathbb{Z}_{\geq 1}}$ (dominated convergence with $\sum f(k) = 2$). Every closed subset of Cantor space is countable or has cardinality $2^{\aleph_0}$; removing countably many finite sets does not change this. Thus for every x the number of set-solutions is countable or exactly $2^{\aleph_0}$.

Remark 5.6 ($x = 2$ is unique as a set). If $\sum_{k \in A} f(k) = 2$, then $\sum_{k \in \mathbb{Z}_{\geq 1} \setminus A} f(k) = 0$, forcing $A = \mathbb{Z}_{\geq 1}$: a unique set-solution. The continuum in Theorem 5.2 lives entirely in the ordering.

Remark 5.7 (Obstruction for rational set-solutions). Any construction in the spirit of Theorem 5.3 needs infinitely many disjoint “swaps” (X, Y) with $\sum_X f = \sum_Y f$. For the resulting x to be rational, these swaps must occur at scales with bounded gaps (otherwise the binary expansion has unbounded zero runs, as in Proposition 5.4). All provable identity families here (Theorems 3.1, Remark 3.3) live at exponentially sparse scales, forced by the same power-of-two rigidity $M + 2 = 2^{M-n+1}$. Producing swaps at every scale is an instance of the exact hitting problem of Section 4. We conjecture a rational x with $2^{\aleph_0}$ set-solutions exists, but do not prove it.

6 Audit

Scope. Proved unconditionally: Lemma 2.1; Theorem 3.1; Remark 3.3; Proposition 4.2; Theorem 4.4 (table entries); Theorem 4.5; Lemma 5.1; Theorems 5.2, 5.3; Proposition 5.4. Stated as open: Open Problem 4.2 and rational set-solutions in Section 5.4.

Computation. All verification in Appendix A uses Python `Fraction` or integer arithmetic only; no floating point.

Compliance. No internet access was used. The investigation is self-contained.

A Exact verification code

The following script verifies Theorem 3.1, Remark 3.3, Table 1, and Theorem 4.5.

```
from fractions import Fraction as F

def f(k):
    return F(k, 2**k)

# Theorem 1:  $n_s = 2^s - s - 1$ 
for s in range(3, 12):
    n = 2**s - s - 1
    blk = list(range(n+1, 2**s - 1))
    assert f(n) == sum(f(k) for k in blk)

# Remark 2: odd  $E \geq 5$ 
for E in (5, 7, 9, 11, 13):
    b = (2**E - 5) // 3 + 1
    a = b + 1 - E
    assert f(a) + f(b) == sum(f(k) for k in range(a+1, b))

# Table certificates ( $t \geq 2$ )
CERTS = {
    1: [4,5,6], 2: [4,5,6], 3: [4,6,8], 4: [5,6],
    5: [6,7,11,13,14], 6: [7,8,11,13,14],
    7: [8,9,12,13,14,15,20,21,24], 9: [10,11,13,14],
    11: [12,13,14], 12: [13,14,15,20,21,24],
    15: [16,17,18,21,22],
}

for n, S in CERTS.items():
    assert len(S) == len(set(S)) >= 2
    assert f(n) == sum(f(k) for k in S)

# Theorem compute: reachability for  $n \leq 2000$ 
def reachable(n, H=250000):
    cur = {n}
    j = n
    while j < H:
        nxt = set()
        for rho in cur:
            r1 = 2 * rho
            if 0 <= r1 <= j + 3:
                nxt.add(r1)
            r2 = 2 * rho - (j + 1)
```

```
        if 0 <= r2 <= j + 3:
            nxt.add(r2)
        j += 1
        cur = nxt
        if 0 in cur:
            return True, j
        if not cur:
            return False, j
    return False, H

for n in range(1, 2001):
    ok, J = reachable(n)
    assert ok, n
print("all checks passed")
```